

4.2 Key Components

4.2.1 Sprout Key Components

Let $\ell_{a_{sk}}$ be as defined in § 5.3 ‘*Constants*’ on p. 77.

Let PRF^{addr} be a *Pseudo Random Function*, instantiated in § 5.4.2 ‘*Pseudo Random Functions*’ on p. 89.

Let $\text{KA}^{\text{Sprout}}$ be a *key agreement scheme*, instantiated in § 5.4.5.1 ‘*Sprout Key Agreement*’ on p. 91.

A new *Sprout spending key* a_{sk} is generated by choosing a *bit sequence* uniformly at random from $\mathbb{B}^{[\ell_{a_{sk}}]}$.

a_{pk} , sk_{enc} and pk_{enc} are derived from a_{sk} as follows:

$$\begin{aligned} a_{pk} &:= \text{PRF}_{a_{sk}}^{\text{addr}}(0) \\ sk_{\text{enc}} &:= \text{KA}^{\text{Sprout}}.\text{FormatPrivate}(\text{PRF}_{a_{sk}}^{\text{addr}}(1)) \\ pk_{\text{enc}} &:= \text{KA}^{\text{Sprout}}.\text{DerivePublic}(sk_{\text{enc}}, \text{KA}^{\text{Sprout}}.\text{Base}). \end{aligned}$$

4.2.2 Sapling Key Components

Let $\ell_{\text{PRFexpand}}$, ℓ_{sk} , $\ell_{\text{ivk}}^{\text{Sapling}}$, ℓ_{ovk} , and ℓ_d be as defined in § 5.3 ‘*Constants*’ on p. 77.

Let $\mathbb{J}^{(r)}$, $\mathbb{J}^{(r)*}$, $\mathbb{J}_{\star}^{(r)}$, $\text{repr}_{\mathbb{J}}$, and $r_{\mathbb{J}}$ be as defined in § 5.4.9.3 ‘*Jubjub*’ on p. 105, and let $\text{FindGroupHash}^{\mathbb{J}^{(r)*}}$ be as defined in § 5.4.9.5 ‘*Group Hash into Jubjub*’ on p. 107.

Let $\text{PRF}^{\text{expand}}$ and $\text{PRF}^{\text{ockSapling}}$, instantiated in § 5.4.2 ‘*Pseudo Random Functions*’ on p. 89, be *Pseudo Random Functions*.

Let $\text{KA}^{\text{Sapling}}$, instantiated in § 5.4.5.3 ‘*Sapling Key Agreement*’ on p. 92, be a *key agreement scheme*.

Let CRH^{ivk} , instantiated in § 5.4.1.5 ‘*CRH^{ivk} Hash Function*’ on p. 80, be a *hash function*.

Let $\text{DiversifyHash}^{\text{Sapling}}$, instantiated in § 5.4.1.6 ‘*DiversifyHash^{Sapling} and DiversifyHash^{Orchard} Hash Functions*’ on p. 81, be a *hash function*.

Let $\text{SpendAuthSig}^{\text{Sapling}}$, instantiated in § 5.4.7.1 ‘*Spend Authorization Signature (Sapling and Orchard)*’ on p. 98, be a *signature scheme with re-randomizable keys*.

Let $\text{LEBS2OSP} : (\ell : \mathbb{N}) \times \mathbb{B}^{[\ell]} \rightarrow \mathbb{B}^{\lceil \text{ceiling}(\ell/8) \rceil}$ and $\text{LEOS2IP} : (\ell : \mathbb{N} \mid \ell \bmod 8 = 0) \times \mathbb{B}^{[\ell/8]} \rightarrow \{0 \dots 2^\ell - 1\}$ be as defined in § 5.1 ‘*Integers, Bit Sequences, and Endianness*’ on p. 76.

Define $\mathcal{H}^{\text{Sapling}} := \text{FindGroupHash}^{\mathbb{J}^{(r)*}}(\text{"Zcash_H_"}, \text{""})$

Define $\text{ToScalar}^{\text{Sapling}}(x : \mathbb{B}^{[\ell_{\text{PRFexpand}}/8]}) := \text{LEOS2IP}_{\ell_{\text{PRFexpand}}}(x) \pmod{r_{\mathbb{J}}}$.

A new *Sapling spending key* sk is generated by choosing a *bit sequence* uniformly at random from $\mathbb{B}^{[\ell_{sk}]}$.

From this *spending key*, the *Spend authorizing key* $ask : \mathbb{F}_{r_{\mathbb{J}}}^*$, the *proof authorizing key* $nsk : \mathbb{F}_{r_{\mathbb{J}}}$, and the *outgoing viewing key* $ovk : \mathbb{B}^{[\ell_{ovk}/8]}$ are derived as follows:

$$\begin{aligned} ask &:= \text{ToScalar}^{\text{Sapling}}(\text{PRF}_{sk}^{\text{expand}}([0])) \\ nsk &:= \text{ToScalar}^{\text{Sapling}}(\text{PRF}_{sk}^{\text{expand}}([1])) \\ ovk &:= \text{truncate}_{(\ell_{ovk}/8)}(\text{PRF}_{sk}^{\text{expand}}([2])) \end{aligned}$$

If $ask = 0$, discard this key and repeat with a new sk .

$ak : \mathbb{J}^{(r)*}$, $nk : \mathbb{J}^{(r)}$, and the *incoming viewing key* $ivk : \{1 \dots 2^{\ell_{ivk}^{\text{Sapling}}} - 1\}$ are then derived as:

$$\begin{aligned} ak &:= \text{SpendAuthSig}^{\text{Sapling}}.\text{DerivePublic}(ask) \\ nk &:= [nsk] \mathcal{H}^{\text{Sapling}} \\ ivk &:= \text{CRH}^{\text{ivk}}(\text{repr}_{\mathbb{J}}(ak), \text{repr}_{\mathbb{J}}(nk)). \end{aligned}$$

If $ivk = 0$, discard this key and repeat with a new sk .